MULTIPLE INTELLIGENCE OF SHS STUDENT OF DFLOMNHS SANBLAS BANGAR, LA UNION SY 2022-2023

A research

Presented to

The faculty of the Senior High School Department of

Doña Francisca Lacsamana de Ortega Memorial

National High School Bangar,La Union

In partial fulfillment

of the requirement in Practical Research 2(Quantitative Research)

By:

Althea C. Belecina

Ken Andrew Muñoz

Vincent Rave Castro

Angelica Damian

Mark Joseph Abanador

Christian Maala

DEPARTMENT OF EDUCATION
REGION 1
DIVISION OF LA UNION
DOÑA FRANCISCA LACSAMANA DE ORTEGA
MEMORIAL NATIONAL HIGH SCHOOL
Bangar, La Union

APPROVAL SHEET

This research entitled

MULTIPLE INTELLIGENCE OF SHS STUDENT OF DFLOMNHS SANBLAS BANGAR, LA UNION SY 2022-2023

prepared and submitted by **ALTHEA CHRISIETTE C. BELECINA, KEN ANDREW MUÑOZ, VINCENT RAVE CASTRO, MARK JOSEPH ABANADOR, ANGELICA DAMIAN,** and **CHRISTIAN MAALA** is hereby recommended for approval and acceptance as a requirement in Practical Research iii.

YVES JILL M. YUKEE, MAT-Math

Adviser

PANEL OF EXAMINERS

Accepted by the Defense Committee on October 30, 2017.

BRIAN RILLORAZA

Chairperson

EDWIN M. LUCINA — Member

REYNA ANN F. PIDOT — Member

Accepted and approved as a requirement in Inquiries, Investigation and Immersion.

GINA M. PEREZ
OIC-SHS Assistant Principal II

JIMA N. ESCOBAR, Ed.D.

Secondary School Principal IV

ABSTRACT

Belecina, A., Muñoz, K., Castro, V., et al. 2022-2023. **MULTIPLE INTELLIGENCE OF SHS STUDENTS OF DFLOMNHS SANBLAS BANGAR, LA UNION.** *Undergraduate of Doña Francisca Lacsamana de Ortega Memorial National High School, La Union Campus, Arts and Design strand, Bangar, La Union.*

Adviser: ***YVES JILL M.YUKEE***

This study looked into the order of multiple intelligences of the students in DFLOMNHS,for school year 2022-2023.Specifically,it sought answers to te following questions: (1) What is the extent of manifestation of the multiple intelligences by the SHS students of DFLOMNHS along; Linguistic; Logical-Mathematical; Musical; Visual-Spatial; Bodily Kinesthetic; Interpersonal; Intrapersonal; and Naturalists intelligences? (2) What is the order of multiple intelligences of the students when grouped according to type of strand and year level? (3) Is there significant difference in the order of multiple intelligences of the students when grouped according to type of strand and year level?

The finding of the study is that the Grade 11 students enrolled most in HE among strand the TVL and Arts and Design track possessing the eight multiple intelligences at varying orders of dominance. The extent of manifestation of different multiple intelligences does not influence by several demographic factors such as the type of strand or Age.

Key words: ; Linguistic; Logical-Mathematical; Musical; Visual-Spatial; Bodily Kinesthetic; Interpersonal; Intrapersonal; and Naturalists intelligences.

TABLE OF CONTENTS

ACKNOWLEDGEMENTS

We would like to thank our subject teacher Ma'am Yves Jill M. Yukee, our Panelists who criticize our study Dr. Brian Rilloraza,our dear panelists, and our Principal Ma'am Jima M. Escobar, for their support and guidance in completing our project on the topic (topic name). It was a great learning experience.

I would like to take this opportunity to express my gratitude to all of my group members Ken Andrew Muñoz, Vincent Rave Castro, Mark joseph Abanador, Angelica Damian, and Christian Maaala. The project would not have been successful without their cooperation and input.

Althea Belecina

Chapter 1
INTRODUCTION

This study looked into the order of multiple intelligences of the students in DFLOMNHS,for school year 2022-2023.Specifically,it sought answers to te following questions: (1) What is the extent of manifestation of the multiple intelligences by the SHS students of DFLOMNHS along; Linguistic; Logical-Mathematical; Musical; Visual-Spatial; Bodily Kinesthetic; Interpersonal; Intrapersonal; and Naturalists intelligences? (2) What is the order of multiple intelligences of the students when grouped according to type of strand and year level? (3) Is there significant difference in the order of multiple intelligences of the students when grouped according to type of strand and year level?

In implementing MI in the classroom,teachers should be able to offer or provide more variety activities which engage students with the topic.Although various learning theories have been adapted in teaching English in the classroom.This is considered as one of the reasons why some learning theories which have been adapted in the clasroon do not work successfully.

We will provide a means of grouping abilities that students posses according to their capabilities into eight comprehensive intelligences:linguistic, logical-mathematical, visual-spatial, bodily kinesthetic, musical, interpersonal, intrapersonal ,and naturalistic. By implying these multiple intelligences,we believe tha teacher could teach students in eight ways and students could learn in many ways.

Multiple Intelligence theory, developed by psychologist Howard Gardner in 1983, suggests that intelligence cannot be measured by a single, general ability, but rather encompasses a variety of distinct abilities or intelligences. According to Gardner, individuals possess different strengths and learning styles, and these intelligences can be nurtured and developed in various ways.

Since the initial proposal, Gardner has suggested the existence of additional intelligences, such as naturalistic intelligence (recognizing and categorizing natural elements) and existential intelligence (contemplating deep philosophical and existential questions).

It is important to note that the theory of multiple intelligences has sparked significant debate and criticism within the field of psychology and education. Some argue that the concept lacks empirical evidence or that the intelligences are more accurately described as talents or abilities rather than true "intelligences." Nonetheless, the theory has had a significant impact on educational practices, encouraging educators to recognize and accommodate diverse learning styles and strengths in their students.

Significance of the study

Specifically, the study will benefit the following;

Curriculum Planners-They early dscovering of the multiple intellgince of the students will guide them in the improvement of instructional programs and policiesin coming up with appropriate teaching materials for quality basic education.

Schools administrators/principals-Having the records of multiple intelligence of students from 1st year to 4th year the principal could make a plan or strategy on how the teachers can train and develop frontliners in the inter-school competiton in oration,debate,math Olympiad,poster-making,sports and athletics and entrepreneurial skills.

Teachers-The output of this study will help them adjust their instructional strategies to the learning modalities of their students.

Students-The students will stay focus on what talents or skills they are going to develop.

Guidance Counselor-If there is an updated record of the multiple intelligence of the students,it could be a basis of giving them on career guidance in line with their field of interest.

Parents-they will be happy to know the multiple intelligence of their children.

The Researchers-They success of this study is a great achievement in the part of the research.

Future Researchers-Results of this tudy will serve as benchmark for future considering the fact that there are still very few researchers who embarked on the study of the multiple intelligence of the student.

Conceptual/Theoretical Framework

The study was guided by several theories and concept;Howard Gardner (1983) developed the theory of multiple intelligence.He proposed eight different intelligences to account for a broads range of human potential in children and adults. These intelligence are:words(LINGUISTIC INTELLIGENCE) (LOGICAL-MATHEMATICAL INTELLIGENCE);social experince (INTERPERSONAL INTELLIGENCE); music (MUSICAL INTELLIGENCE) and nature (NATURAL INTELLIGENCE).

One of the most remarkable features of the theory of multiple intelligence how it provides eight different potential pathways to learning. If a teacher is having difficulty reaching a student in the more traditional linguistic of logical ways of instruction, the theory of multiple intelligences suggest several other ways in which the material might be prevented to facilitate effective learning.

He doesn't have to teach or learn something in all eight ways, just see what the possibilities are, and then decide which particular pathways interest him most or seem effective teaching or learning tools. The theory of multiple intelligences is so intriguing because it expands our horizon of variable teaching and learning tools beyond the conventional linguistic and logical methods used in most school in most school such as lectures,textbooks writing assignments and others

"the disciple of active involvement clearly" elucidates the significance of the kinesthetic intelligence in the part of the student. The movement of the different parts of the body as he actively participates in class or in school definitely sharpens his minds and enhances his learning ability.

This is in consonance with the principle of self-activity by Frobel as quoted by Garo (2008) which states that learning is the result of the activity of the child himself.

The student should not be made to watch or to observe what is the student should not be made to watch or to observe what is happening. He sould not contribute his ideas or concepts to solve issue.

One cannot learn to solve mathematical problems unless he tries to undergo the process himself.

The principle of discovery-builds confidence in the part of the students for being able to discover his strongest learning style based on the five stimuli such as environmental, emotional, sociological. physiological and psychological (DUNN, 1993) The discovering of his perceptual strengths lead to the development of his dominant multiple intelligence.

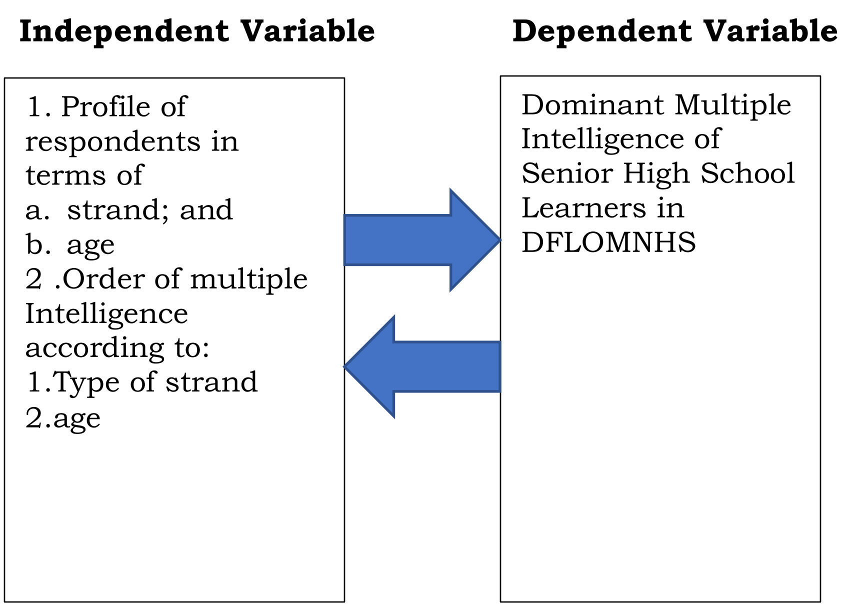
With the principle of discovery, the kind of talent and skill. The student wishes to presue in greatly enhance.

The principle of socialization is highly related to the interpersonal intelligence of the student. He must mingle with people in the society in order to acquire good habits, attitudes, values, knowledge and skills, which are very necessary for the improvement of his personality.thus,according to essentialism, the school must be structured as a miniature society where students learn the of living effectively in one own milieu.

In line with the aforementioned theories and concepts, this study was conceived and presented in the form of a paradigm.

Research Paradigm

Figure 1



Definition of terms

Multiple Intelligence- Different level of intelligence possessed by even individual which includes verbal linguistics, logical, mathematical, musical, visual-spatial ,bodily kinesthetic, interpersonal, intrapersonal, and naturalistic.

Linguistic intelligence- Students with linguistic intelligence have the ability to use language and words in various forms.

Logical intelligence - Intelligence-Students with logical mathematical intelligence can think logically and systematically and solve problems rationally

Musical Intelligence-students with musical intelligence are exceptionally bright at rhythmic pattern,pitch,timbre melody and tone.

Visual-Spatial Intelligence-students with visual-spatial intelligence are able to think in three dimensions and can be create mental image.

Bodily-kinestetic Intelligence-students with bodily-kinestetic intelligence we their body to communicate with others.

Inter-Personal Intelligence-It means relationship between persons.Students with interpersonal intelligence can relate with people very well

Intra-Personal Intelligence-Students who process intra-persnonal intelligence are sensitive to their own feelings moods and mind

Natural Intelligence-students with naturalist intelligence are very sensitive to the natural world

Statement of the Problem

This study aims to identify the dominant multiple intelligence of SHS student of DFLOMNHS school year 2022-2023

Specifically, it sought to answer the following questions;

1.What is the profile of the respondents in terms of:

a.strand;and

b.age?

2.What is the extent manifestation of the following multiple intelligence by SHS students of DFLOMNHS

a.Linguistic intelligence

b.Logical-Mathematical intelligence

c.Musical intelligence

d.Visual-Spatial intelligence

e.Interpersonal intelligence

f. Intrapersonal intelligence

g.Naturalist intelligence

3.What is the order of multiple intelligence of the students of the students when grouped according to;

a.type of strand

b.age

4.Is there a significant difference in the order of dominance of multiple intelligence of students when grouped according to;

a.type of strand;and

b.age

Statement of the Hypothesis

There is no significant difference in the order of dominance of intelligence of students when grouped according to strand and their age.

Chapter 2

Research Methodology

This chapter categorically presents the research design, instrumentation and data,local of the study, instrumentation and data collection,and the tools for data analysis.

Research Design

This study made use of descriptive comparative method since it describes and compares the dominant multiple intelligences of the SHS students of DFLOMNHS

According to Best (2008),the descriptive method involves describing,analyzing,interpreting conditions that exist.It also includes some types of comparisons,contrast,and attempt to discover relationship between existing non-manipulative variables. It concerned with the conditions of relationship that exist,opinions that are held, processes that are going on,or trends that are for development.

Research environment

This study was conducted in DFLOMNHS San Blas, Bangar, La Union.

The school main campus has special programs for students with high aptitude in science,arts, or sports.

Bangar Municipal High School opened in 1972,on initiative of former Mayor Avelino Pascua.It started with two classrooms for first and secondary year levels, at the old Gabaldon Campus. The school was later renamed as Dona Francisca Lacsamana de Ortega Memorial High School. The Casa Cristo Annex was added in 1993. In 1994, the school become the Mother Institution of the Regional Science High school for Region 1, after it was recognized by the department of Education, Culture and ports as one of the "most effective secondary schools in the Philippines."

Respondent

This study involved 86 of the 110 senior high school students DFLOMNHS San Blas, Bangar, La Union for school year 2022-2023.

The adopted principle of random sampling was sample size determined using the formula by Slovin (1960).

5% was the applied error of margin to each strands with the total of 86 respondents of SHS students.The students were distributed as follows:

Frequency of Distribution

P=Population S=Sample

	HE	AP	ICT	TOTAL
POPULATION	79	11	20	110
SAMPLE	63	7	16	86

Data Gathering tool

The data gathering tool used in this study is the questionnaire consisted of 40 items as indicators for the multiple intelligences of the respondents. It is a likert type of questionnaire with five(5) choice. A Likert scale is a psychometric scale commonly involved in research that employs questionnaires. It is the most widely used approach to scaling responses in survey research, such that the term is often used interchangeably with rating scale, although there are other types of rating scales.

The questionnaire is based on the Howard Gardner theory.The theory of multiple intelligences suggests that the traditional notion of intelligence,based on I.Q. testing,is far too limited.

Data Gathering Procedure

Prior to the administration of the questionnaire to the subject-respondents, we will make sure to secure a permit to the school administrator or the principal of DFLOMNHS to take a sample on SHS and SPA students for our study.

After the valid consent for gathering the data,the finding of sample will start,we'll make sure to know how they feel about the study and acknowledge it, in other words students who's willing to answer our questions in the questionnaire will serve as our respondents.

To be sure of a one hundred percent retrieval, we will personally administer the questionnaire with the ad of the class advisors.

After a retrieval,the set of questionnaire will then be coded ready for data processing.

Statistical treatment of data

The statistical tool that was utilized in the treatment of data for problem number (1) on the multiple intelligences of SHS students of DFLOMNHS on every strand and year level were weighted mean and ranking. The computation of the mean value will be done by multiplying the raw scores by the numerical weight in each column on 5-4-3-2-1 scale of values, adding the products obtained and dividing the sum by the total replies for a specific item.

For the problem number two (2) if there are significant differences in the order of intelligences when analyzed into type of strand and year level, z-test will be used.

For problem number three (3) the one way analysis was used to determine significant difference.The formula for ANOVA is (Pagoso,1992)

Chapter 3

RESULTS AND DISCUSSION

The data that are presented, analyzed and interpreted in this chapter are culled out from the questionnaire answered by the respondents. The arrangements are logically based on the order of statement of the problems.

3.1 Profile of the Respondents

3.1.1 Strand

Table 3.1 shows the percentage distribution of the grade 11 students of DFLOMNHS respectively to their strand, the highest number of respondents came from the learners of Home Economics (HE) strand with the frequency of 62 taking mostly as our respondents to the study with 72% percent, followed by the strand ICT with a frequency of 17 taking 20% of our study as our respondent, and AP strand taking the last rank of having the frequency of 7 and percentage of 8%. The table shows that, among the strands chosen to this study ,students wanted to learn most about Home Economics, but we won’t focus on that to our study.

Most of our respondents were from the strand of HE, they want to have a knowledge of how to cook and manage aspects of a household through a class. Not only does home economics teach students about cooking and safety but it also builds responsibility. It teaches teens to use the techniques they learned in class in their home life. When teens learn how to take care of their household and themselves, it helps them to become more responsible at home.

Although many may think that a class isn’t needed to teach about the basic necessities, having a class solely devoted to these aspects is exactly what we need in school. Home economic classes weren’t taught until the early 20th century. An example that had the most impact was the “Treatise on Domestic Economy for the Use of Young Ladies at Home,” written by Catharine Beecher. Beecher argued that it was important to teach domestic life and apply scientific principles to things like cooking and housekeeping. Home economic classes would help teens develop skills in family financing, nutrition, cooking and other various skills for life.

Same as Ict and Ap strand, today information and communication technologies are the one thing and so the repertoire of technologies expands further to encompass computers and computer-related products, email, MMS, and other forms of communication (Finger et al., 2007). If the learners put a lot of thought into their planning, a higher degree of engagement will be notice and this can lead to the development of 21st-century skills such as complex thinking, creative problem-solving, and collaboration.

In life ‘knowing how’ is just as important as ‘knowing that’. Art, craft and design introduces participants to a range of intellectual and practical skills. It enables learners to use and understand the properties of a wide range of tools, machines, materials and systems. It provides children, young people and life-long learners with regular opportunities to think imaginatively and creatively and develop confidence in other subjects and life skills.

Strands were divided to see if different skills of the grade 11 students affect their multiple intelligence hence the significant difference between the strands.

Table 3.1.1 Frequency distribution of respondents' strand

STRAND	FREQUENCY	PERCENT
AP	7	8%
HE	62	72%
ICT	17	20%
TOTAL	86	

3.1.2 Age

In this table, it shows the distribution and the percentage of every ages recorded of grade 11 students of DFLOMNHS, and it shows that the average age from the range of 15-20 is 16 years old. According to the table, The range age 15-17 has a frequency of 73 and a percent of 85%,and for the range of age 18-20 has only containing of 13 respondents with 15%,the total of the grouped age was 86 respondents.

In the Philippines, Grade 11 (Filipino: Baitang Labingisa) is the first year of Senior High School and the fifth year of High School curriculum.

Topics discussed depend on the four tracks and their strands. since school year 2016–2017. Grade 11 students are usually 16–17 years old.

In our study, the age recorded was with the range of 15-20. Teenagers (15-17 years of age), This is a time of changes for how teenagers think, feel, and interact with others, and how their bodies grow. Most girls will be physically mature by now, and most will have completed puberty. Children in this age group might learn more defined work habits, show more concern about future school and work plans, be better able to give reasons for their own choices, including about what is right or wrong. 18- to 20-year-olds, who much prefer to be called "young adults." This is the age where physically the growth and development has slowed, but socially and emotionally they are transitioning from what has been somewhat of a routine and protective environment to the unknown.

Problem behavior was related to lower crystallized intelligence, whereas psychological maturity was related to higher crystallized intelligence and better performance . Psychosocially mature adolescents had significantly higher composite IQ scores than did pseudo mature adolescents. Mature adolescents also showed advantages in crystallized and fluid intelligence, and in performance , compared to their less mature counterparts. Therefore, maturity might affect their multiple intelligence.

Table 3.1.2 Age distribution of respondents

Age	Frequency	Percentage	Mean
15-17	73	85%	16
18-20	13	15%	
Total:	86	100%	

3.2 Extent Manifestation of the Multiple Intelligence of the Students

After knowing the profile of the SHS students, The first problem of the study asks the extent of manifestation of the multiple intelligences among the SHS students of DFLOMNHS grade 11.

Table 3.3 reflects the extent manifestation of the multiple intelligences among grade 11 students of DFLOMNHS. The most manifested was naturalists intelligence with mean value of 4.07 described as often manifested and the least manifested was logical/mathematical with a mean of 2.59 described as seldom manifested. The extent manifestation in descending order with their mean values were: Naturalists (4.07),Interpersonal (3.63),Visual-Spatial (3.45),Intrapersonal (3.43),Musical (3.28),Bodily kinesthetic (3.11) tied with Linguistic (3.11),and Logical-Mathematical (2.59).

The high level of naturalists intelligence of the grade 11 students in DFLOMNHS was a manifestation of their love for plants, trees and flowers. They collect seashells and fishes and display them in an aquarium. They are conscious of garbage and waste disposal. They are excited bout biological museum tours. They are interested in scientific experiments, especially those relating to life science. They prefer animal and birds for pet keeping. They are delighted by natural settings or phenomena.

According to Lee (1991),the low indication to logical-mathematical intelligence by the students was attributed to three main factors fear of the subject, misconception of its allegedly "impractical" nature, and refusal to follow the rules of it's logic).It was also due to lack of practice to the four basic fundamental operations in mathematicians, such as: addition, subtraction, division and multiplication.

On the other hand the low manifestation of logical-mathematical intelligence was attributed to the fact that these students are naturally playful as they are nature lovers. They manifest low concentration in activities like solving mathematical problems and reasoning.

Students with naturalists intelligence should be inspired to become botanists; biochemists; archeologists; aquatic researchers; zoologists; veterinarians; and biologists

Table 3.2 Extent of manifestation of the multiple intelligence of the respondents

Multilple Intelligences	Mean	Descriptive Equivalent	Rank
Linguistic Intelligence	3.11	SO	6.5
Logical/Mathematical Intelligence	2.59	S	8
Musical Intelligence	3.28	SO	5
Visual/Spatial Intelligence	3.45	O	3
Bodily Kinesthetic Intelligence	3.11	SO	6.5
Intrapersonal Intelligence	3.43	O	4
Interpersonal Intelligence	3.63	O	2
Naturalists Intelligence	4.07	O	1

3.3 Order of Multiple Intelligence of Respondents

Tables 3.3.1 and 3.3.2 illustrates the order of multiple intelligence of the students when grouped according to their strand they are in and their age.IT categorically answers problem number three (3).

3.3.1 Strand

Table 3.3.1 shows that the most dominant intelligence possessed by the students from AP was “naturalistic” with mean value of 4.24 and the least intelligence manifested by them was “logical mathematical” with a mean value of 2.89.The order of other multiple intelligences in descending order with their mean values are; Visual-spatial (3.93),Linguistic(3.54), Intrapersonal (3.52), Musical (3.51), Interpersonal (3.31), Bodily kinesthetic (3.26).

The most dominant intelligence possessed by the learners of ICT was also “naturalistic” with a mean value of 4.37 and the least intelligence manifested by them was “logical/mathematical” with a mean value of 2.51.The order of other multiple intelligence in descending order and their mean; Intrapersonal (3.80), Musical (3.29) and Visual-spatial (3.29), Bodily kinesthetic (3.20), Interpersonal (2.96), Linguistic (2.79).

The highest rank of intelligence possessed by the learners of HE was also “naturalistic” with a mean value of 3.96 and the least intelligence manifested by them was “logical mathematical” with a mean of 2.57. The order of other multiple intelligence in descending order and their mean; Intrapersonal (3.59), Visual-Spatial (3.44), Interpersonal (3.59), Musical (3.25),Linguistic (2.57), Bodily kinesthetic (3.07).

The findings imply that the learners of AP,ICT,and HE are lovrers of nature. This is manifested by their great interest to stay under the shade of trees inside the school campus during their free time. Further, it is observed that the students were eager to join fieldtrips and nature tripping.This characteristics of the students justifies the dominance of naturalistic intelligence in them.

However, the students’ low level of achievement in mathematics particularly in problem solving proves that the least intelligence present in them is the logical mathematical Intelligence.

The other leading intelligence of Ap learners were;visual-spatial (3.93),and Linguistic (3.54),the visual spatial intelligence of the Ap learners shows that they could create simple designs,patterns and shape.They have the ability to reproduce scenes and objects through painting ,sculting or constructing small replica of houses.The linguistic intelligence of Ap learners,involves sensitivity to spoken and written language,the ability to learn languages and the capacity to use language to accomplish certain goals.

The top 3 next to Naturalistic intelligence of ICT students were;Intrapersonal (3.80),and musical tying with visual spatial (3.29). The students of ICT likes to work independently after instructions are given,they have the ability to control their emotions,feelings and moods.The musical intelligence among the students shows that they can comprehend musical symbols easily.They have the ability to produce a variety of songs and express their idea through music,these is according to Tenedero (1998).

The intelligence next to naturalistic intelligence of HE students were;Intrapersonal (3.59),and visual-spatial (3.44).they have the capacity to understand oneself to appreciate one's feelings ,fears and motivation in Howard Gardners (1993) view.They have the potential to recognize and use patterns of wide space and more confined areas.

Though the mentioned intelligences of the learners of each strand were just the next leads to naturalistic intelligence,that means it's not their best skills among the intelligences.

Table 3.3.1 Order of multiple intelligence of grade 11 students per strand

Multiple Intelligence	AP		HE		ICT	
	MEAN	RANK	MEAN	RANK	MEAN	RANK
Linguistic	3.54	3	2.79	7	3.15	6
Logical-Mathematical	2.89	8	2.51	8	2.57	8
Musical	3.51	5	3.29	3.5	3.25	5
Visual Spatial	3.93	2	3.29	3.5	3.44	3
Bodily kinesthetic	3.26	7	3.20	5	3.07	7
Interpersonal	3.31	6	2.96	6	3.30	4
Intrapersonal	3.52	4	3.80	2	3.59	2
Naturalistic	4.24	1	4.37	1	3.96	1

3.3.2 Age

Table 3.3.2 presents the order of multiple Intelligence of then respondents arranged according to age.

It can be gleaned from table 3.5 that the foremost intelligence among the age 15-20 was "naturalist" with a mean of 4.07 from the age 15-17 and 4.04 from the age 18-20,and the lowest is logical/mathematical in both range of ages, with the mean of 2.57 for 15-17 and, 2.68 for 18-20.

The findings imply that the students within the age 15-20 are lovers of nature. This is manifested by their great interest to stay under the shade of trees inside the school campus during their free time. Further, it is observed that the students were eager to join fieldtrips and nature tripping.This characteristics of the students justifies the dominance of naturalistic intelligence in them.

The intelligences between the lowest and highest ranked intelligence of the students within age 15-17 were;Interpersonal (3.64),Intrapersonal (3.62),Visual-spatial (3.47),Musical (3.29),Linguistic (3.07),and Bodily kinesthetic (3.05).Whithin the age 18-20 were;Intrapersonal (3.69),Bodily kinesthetic (3.45),Visual spatial tying wiyh Linguistic (3.33),Interpersonal (3.20),and Musical (3.23).

The top 2 leading intelligences of students ages 15-17 next to naturalistist were;,Interpersonal (3.64) and, Intrapersonal (3.62) respectively. The high ranking of Interpersonal intelligence of students aging 15-17 shows that they like to do things or perform activities and are highly motivated,when paired with friends. They can also work independently after instructions are given,they have the ability to control their emotions,feelings and moods,These justifies that Intapersonal was one of most intelligence of the students within the age 15-17.

The next leading intelligeces of students within the age of 18-20 were; Intrapersonal (3.69), and Bodily kinesthetic (3.45).The Intrapersonal intelligence being the next leading intelligence means that,they know how to ggive meaning to personal likes and dislikes, they have a good sense of concentration,and they attempt to seek out and understand inner feelings and thoughts.Bodily kinesthetic is the next lead to Intrapersonal,,it shows their love to dance, join sports and scouting,become members of dramatic club,and other cultural activities.

Teachers could enhance the naturalists intelligence of students by initiating tree-planting activities in school or immediate environment.They would let their students bring plants and animals as specimen in Biology class. They will visit botanical gardens and zoological museums.They will teach their students on how to grow plants and learn how to crop,graft or pollinate them.They will take time to observe marine life and lakes,rivers,beaches or the sea.

The low level achievement of mathematics of the students could still be improved.theachers should encourage them to to work on puzzles,tell them to watch television program featuring scientific concepts,like in the discovery channel.They should establish or lead a math or science club,in order to create a study group to discuss recent scientific discoveries.

Table 3.3.2 Order of multiple intelligence of student according to age

Multiple Intelligences	Age 15-17		Age 18-20	
	Mean	Rank	Mean	Rank
Linguistic	3.07	6	3.33	4.5
Logical-Mathematical	2.57	8	2.68	8
Musical	3.29	5	3.23	7
Visual-spatial	3.47	4	3.33	4.5
Bodily kinesthetic	3.05	7	3.45	3
Interpersonal	3.64	2	3.20	6
Intrapersonal	3.62	3	3.69	2
Naturalistic	4.07	1	4.04	1

3.4 Significant difference of Multiple Intelligence of the Students and their Profile

Table 3.4 Presents the difference of the order of manifestation of the multiple intelligences of the students when grouped according to strand age. We have statistically significant evidence at $\alpha=0.05$. The hypothesis is that there is no significant difference to the order of multiple intelligence when grouped according to strand and age.

The scores of linguistic intelligence of students difference between the strands is significant. We reject H_0 (null hypothesis)because the p-value of 0.05 is equal to the significant evidence of 0.05. Comparing by their age, there is no significant difference to the possession of linguistic intelligence of the students, because the p-value of 0.33 is greater than the significant evidence of 0.05.

The scores of logical/mathematical intelligence of students difference between the strands is not significant. We accept H_0 (null hypothesis)because the p-value of 0.42 is greater than the significant evidence of 0.05.Comparing by their age, there is no significant difference to the possession students of logical/mathematical intelligence of the, because the p-value of 0.57 is greater than the significant evidence of 0.05.

The scores of musical intelligence of students difference between the strands is not significant. We accept H_0 (null hypothesis)because the p-value of 0.68 is greater than the significant evidence of 0.05.Comparing by their age, there is no significant difference to the possession of musical intelligence of the students, because the p-value of 0.87 is greater than the significant evidence of 0.05.

The scores of visual-spatial intelligence of students difference between the strands is not significant. We accept H_0 (null hypothesis)because the p-value of 0.36 is greater than the significant evidence of 0.05.Comparing by their age, there is no significant difference to the possession of visual-spatial intelligence of the students, because the p-value of 0.65 is greater than the significant evidence of 0.05.

The scores of bodily kinesthetic intelligence of students difference between the strands is not significant. We accept H_0 (null hypothesis)because the p-value of 0.71 is greater than the significant evidence of 0.05. Comparing by their age, there is a significant difference to the possession of bodily-kinesthetic intelligence of the students, because the p-value of 0.05 is equal to the significant evidence of 0.05.

The scores of interpersonal intelligence of students' difference between the strands is significant. We reject H_0 (null hypothesis)because the p-value of 0 is less than the significant evidence of 0.05. Comparing by their age, there is no significant difference in the possession of visual-spatial intelligence of the students, because the p-value of 0.19 is greater than the significant evidence of 0.05.

The scores of intrapersonal intelligence of students' difference between the strands is not significant. We accept H_0 (null hypothesis) because the p-value of 0.97 is greater than the significant evidence of 0.05. Comparing by their age, there is no significant difference to the possession of intrapersonal intelligence of the students, because the p-value of 0.78 is greater than the significant evidence of 0.05.

The scores of naturalistic intelligence of students' difference between the strands is significant. We reject H_0 (null hypothesis)because the p-value of 0.04 is less than the significant evidence of 0.05. Comparing by their age, there is no significant difference to the possession of naturalistic intelligence of the students, because the p-value of 0.83 is greater than the significant evidence of 0.05.

In general, the order of multiple intelligence of students does not differ significantly when grouped according to strand and their age

This means that the type of strand the students went in and their age does not affect the order of their multiple intelligences. Whether the students are enrolled with the strand of Ap, Ict, or He, or if they're older or younger than the designated age for grade 11, the orders of their multiple intelligences do not differ.

Table 3.4 Order of multiple intelligence and their profile.

MI	Profile		Mean	P-value	Remarks
Linguistic	Strand	AP	3.54	0.05	Significant
		HE	3.15		
		ICT	2.79		
	Age	15-17	3.07	0.33	Not Significant
		18-20	3.33		
Logical/ Mathematical	Strand	AP	2.89	0.42	Not Significant
		HE	2.57		
		ICT	2.51		
	Age	15-17	2.57	0.57	Not Significant
		18-20	2.68		
Musical	Strand	AP	3.51	0.68	Not Significant
		HE	3.25		
		ICT	3.29		
	Age	15-17	3.29	0.87	Not Significant
		18-20	3.23		
Visual-Spatial	Strand	AP	3.93	0.36	Not Significant
		HE	3.44		
		ICT	3.29		
	Age	15-17	3.47	0.65	Not significant
		18-20	3.33		
Bodily Kinesthetic	Strand	AP	3.26	0.71	Not Significant
		HE	3.07		
		ICT	3.2		
	Age	15-17	3.05	0.05	Significant
		18-20	3.45		
Interpersonal	Strand	AP	3.31	0	Significant
		HE	3.57		
		ICT	2.96		
	Age	15-17	3.46	0.19	Not Significant
		18-20	3.28		
Intrapersonal	Strand	AP	3.74	0.97	Not Significant
		HE	3.81		
		ICT	3.8		
	Age	15-17	3.62	0.78	Not Significant
		18-20	3.69		
Naturalistic	Strand	AP	4.24	0.04	Significant
		HE	3.96		
		ICT	4.37		
	Age	15-17	4.07	0.83	Not Significant
		18-20	4.03		

Chapter 4

This chapter consists of conclusions and recommendations based on the gathered data.

Conclusions

1. The average age of our respondent is 16 meaning our respondents is mostly teenagers, and Grade 11 students enrolled most in HE among strand AP, HE, and ICT.

2. The students of Grade 11 AP, HE, and ICT of DFLOMNHS possess the eight multiple intelligences at varying orders of dominance.

3. The most dominant intelligence of grade 11 students of DFLOMNHS is naturalistic and the least is logical/mathematical either grouped to their strand or their age.

4. The extent of manifestation of different multiple intelligences does not influence by several demographic factors such as the type of strand including AP, HE, and ICT, or Age within the range of 15-20.

Recommendations

As a result of the conclusions drawn from this study, the following are hereby recommended:

1.The teachers may consider the findings of this research as a basis for reinforcing the multiple intelligence of the students and crafting their instructional materials/strategies to suit and improve the multiple intelligences of the students.

2.The guidance counselors must establish a record of the multiple intelligences of the students to strengthen their insights in giving career guidance to them.

3.The parents must encourage their children to pursue the career that suits their capacity and intelligence and not according to their desire.

4.The students must have the passion to develop their special talents and dominant intelligences for their own advantage. They should likewise improve their least dominant intelligence.

5.Teachers and parents could enhance the intrapersonal intelligence of the students by encouraging them to read self-motivation books, like biographies of famous personalities. Teachers and parents can likewise enhance bodily-kinesthetic intelligence.

References:

R Gardner, Howard(1995**) Multiple Intelligences:The Theory:Basic Books:Gardner (1995) Leading Minds,An Anatomy of Leadership**, New York;Basic Books.

Dunn,Rita and Kenneth Dunn(1993) **Teaching Secondary Students Through their Individual Learning Styles**.Practical Approaches for grades 7-12 MA:Ally and Bacon

Garo,Candelaria,P.(2008)**Theories and Principles of Educational Tecgnology**.National Bookstore.

Pagoso,Cristobal,(2005)**Statistics for social Sciences**.Manila Philippines.Rex Bookstore.

Best,(2008),Stated that the descriptive method involves describing,analyzing,interpreting conditions that exist.It also includes some types of comparisons,contrast,and attempt to discover relationship between existing non-manipulative variables.It concerned with the conditions of relationship that exist,opinions that are held,processes that are going on,or trends that are for development.

Lee,Queenna,N. (1991).*Why Be Afraid of Math.*Bookmark, Inc. Makati City.

Gardner, Howard(1995**) Multiple Intelligences:The Theory:Basic Books:Gardner (1995) Leading Minds,An Anatomy of Leadership**, New York;Basic Books.

https://dailygazette.com/2018/06/01/students-learn-important-life-skills-in-home

economics/?fbclid=IwAR3fh7czwVY7Njm9zbPLoPjXHAhoGlsH2kXavnkKgm2rm5EoT4GslvHQCK8#:~:text=Not%20only%20does%20home%20economics,become%20more%20responsible%20at%20home

Appendix A

Letter to respondents

Dear Respondents,

In line with our study multiple intelligences of SHS students Of DFLOMNHS,we need your cooperation in patiently answering with utmost honesty all the items..This questionnaire intented to collect data about the multiple intelligences of SHS students to identify it's extent manifestation,order,and significant difference between strands for academic purposes only.Our collection of data will be kept confidential.Thank you for your cooperation.

Appendix B

Questionnaire

Description : 5-always 4-often 3-sometimes 2-seldom 1-never

Indicators	Response				
	5	4	3	2	1
1.Loves to build Vocabulary.					
2.Solves the puzzle easily.					
3.Loves to sing and hum.					
4.Likes to sketch or draw posters.					
5.Skillful dancer.					
6.works well with people.					
7.Pursue personal interest.					
8.Takes a good care off domestic animals.					
9.Talks clearly and convincingly.					
10.Loves to solve problem.					
11.Enjoy writing songs.					
12.Loves the beauty of nature.					
13.Play sports and athletics.					
14.Makes other feels important.					
15.Controls his/her own feelings.					
16.Is concious of garbage Disposal.					
17.writes interesting stories.					
18.Likes abstract formula.					

19.Loves beating or conducting.					
20.Responds well to physical stimulus.					
21. establishes good eye contact.					
22.Manages his/her personal growth.					
23.Manages his/her personal growth.					
24.Cares for plants,trees,and flowers.					
25.Fun of spelling and talking.					
26.Works with number with ease.					
27.Remembers melodies.					
28.Loves color and design.					
29.Enjoys physical execise.					
30.Is highly sociable.					
31.Strives for self interest.					
32.Is attracted by natural settings.					
33.Loves to recite poems.					
34.Joins math contest.					
35.Enjoys listening musical sound.					
36.Appreciate beauty spots.					
37.Uses body language.					
38.Enjoys the company of others.					
39.Ask about the “why” of things					
40.Loves to study plants and animals.					

Appendix C

Letter to Principal

JIMA N. ESCOBAR, EdD
Secondary School Principal IV
DFLOMNHS
Bangar, La Union

Madam:

Greetings

In partial fulfillment of the requirements of the grade 12 learners in the subject Research Project (Inquries and Immersion) they have to conduct a research.

In view of this manner, we would like to request permission from your good office to allow the students to conduct interviews, float research questionnaires and collect data from the identified respondents as part of data collection of their research. Rest assured that all the data gathered will be kept in utmost confidentially.

Thank you for your positive response to this matter.

Respectfully yours,

YVES JILL M. YUKEE
Subject Teacher

REYNA ANN F. PIDOT
Subject Teacher

Recommending Approval:

GINA M. PEREZ, MAEd
HT-III, MAPEH and
OIC-Asst. Principal II, SHS

Approved:

JIMA N. ESCOBAR, EdD
Secondary School Principal IV

Appendix D

Sample statistical Computations

Weighted Mean Formula

$$\bar{x} = \frac{\sum w_n x_n}{\sum w_n}$$

The computation of the mean value will be done by multiplying the raw scores by the numerical weight in each column on 5-4-3-2-1 scale of values, adding the products obtained and dividing the sum by the total replies for a specific item.

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Where

Z=z-test
X1=sample mean of the first group
X2=sample mean of the second group
μ1= population mean of the first group
μ2=population mean of the second group
σ1=standard deviation of the first group
σ2=standard deviation of the second group
n1=sample size of the first group
n2=sample size of the second group

$$F = \frac{Mssb}{Mss}$$

Where F=f-value

Mssb=mean of squares between column

Mss=mean of squares within column

Curriculum Vitae

A. Personal Data

Name :Althea Chrisiette C. Belecina

Birthday:August 24,2005

Birthplace:Central west #2

Father: Christopher O. Belecina

Mother: Ivina C. Belecina

Home address: Sanblas Bangar La Union

B. Educational background

1.Elementary:Bangar Central School

2.Highschool (junior high): DFLOMNHS

3.Highschool (senior high): DFLOMNHS

A. Personal Data

Name:Ken Andrew Muñoz

Birthday:march 3 2004

Birthplace:BayBay leyte Father: Rodolfo Muñoz

Mother: Fe Muñoz

Home address:Central East 2

B.Educational Background

1.Elementary:RUS ELEMENTARY SCHOOL

2.Highschool (junior high): DFLOMNHS

3.Highschool (senior high) : DFLOMNHS

A. Personal Data

Name: Vincent Rave Castro

Birthday: June 05, 2004

Birthplace: Tagudin Ilocos Sur

Father: Dennis Castro

Mother: Mardy Castro

Home address: Ubbog

B.Educational background

1.Elementary: BANGAR CENTRAL ELEMENTARY SCHOOL

2.Highschool (junior high): DFLOMNHS

3.Highschool (senior high) : DFLOMNHS

35

A. Personal Data

Name:Anjielica Damian

Birthday:May30 2005

Birthplace:Ma.Cristina East Bangar Launion

Father:Alejo Elpidio Damian

Mother:Janet Damian

Home address:Ma. Cristina East bangar launion

B.Educational background

1.Elementary:Ma.Cristina East

2.Highschool (junior high): DFLOMNHS

3.Highschool (senior high) : DFLOMNHS

A. Personal Data

Name:Mark joseph Abanador

Birthday:Dec,04,2003

Birthplace:tiaong quezon

Father:Salvador Abanador

Mother:Evilyn Abanador

Home address:Barraca,Bangar La Union

B.Educational background

1.Elementary: Caggao ES

2.Highschool (junior high): DFLOMNHS

3.Highschool (senior high): DFLOMNHS